Matihus Molina Larios

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Professional Profile

I am a final-year mathematics student at National University of San Marcos with interest in Ergodic Theory, Mathematical applications to health, and Formally verified mathematics. Recently, I have developed a particular interest in the formalization of mathematics using LEAN4. I am also involved in the preparation of research papers related to machine learning applied to lung cancer and a comparative study of numerical methods for differential equations of Lane-Emden type. My current objective is to strengthen my training in pure and applied mathematics and contribute to the development of the academic community in Peru. After graduating, I aspire to pursue postgraduate studies abroad exploring different areas of mathematics.

Education

Universidad Nacional Mayor de San Marcos **Lima, Perú**
B.Sc. Student in Mathematical Sciences *03/2022 – 12/2026 (expected)*
Cumulative (CGPA): 15.19/20

Cultural Center LIDEMPERÚ **Lima, Perú**
Graduate , English Program *2018 – 2021*

Languages

Spanish: Native

English: Intermediate (B1)

Portuguese: Basic (A2)

Academic Projects

PICV-UNMSM
Proyect: "A First Encounter with Chaotic Dynamical Systems" *01/2025 – 03/2025*

- Academic advisor: Dr. Jorge L. Crisóstomo
- Study of topological conjugacy between Q_c and the Bernoulli Shift.
- Aplication of concepts from differential equation,funcional analysis,topology,and geometry.
- Click here

Stellar Structure Modeling
Numerical solution for differential equations of Lane-Emden type *10/2025 –12/2025*

- Analysis of decomposition methods and their differences.
- Numerical implementation and evaluation of the efficienty of these methods.
- Click here

Academic Events

E′(UBB) = 2025
UBB- Concepción *October 2025*
My first international academic experience. I greatly enjoyed the city and the opportunity to learn about a different culture. This experience motivated my interest in pursuing a master's degree abroad.

VII EICM
UNASAM- Huaráz *December 2025*
This was my first experience presenting a research project at an event of this scale. The project presented focused on stellar structure modeling via Lane–Emden type equations.

Summer School and Workshop on Optimization and Operator Learning
UNT – Trujillo *Febrary 2026*
I participated in this important event, putting my knowledge of optimization theory and programming to the test.

Research Interests

Pure Mathematics:
- Functional analysis.
- Ergodic theory.
- Mathematical foundations of artificial intelligence.
- Formally Verified Mathematics.
- Mathematical applications to human health.

Technologies:
- Programming languages: Python, R, MATLAB,LEAN4.
- Libraries: NumPy,SciPy,Pandas, Scikit-Learn, Matplotlib,PyTorch.
- Tools: Git, LaTeX.

Grants, Distinctions

BECA PERMANENCIA 2025 ,PRONABEC

Recognition from my faculty for my ACADEMIC EXCELLENCE

Extracurricular Activities

Co-Director of the IEEE Computational Intelligence Society project area – UNMSM.

Former member of the cultural and artistic salsa group SALSUNI – UNMSM.

Former member of the university judo team – UNMSM

Active member of the BILDHEIT.

Certificates

Certificates of coursework,academic achievements are available in the following repository.